Type 2 Diabetes Digital Twin

Forecast glucose 2 hours ahead. Warn early. Test remedies.

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Happiest Health Digital Twin Challenge 2026

github.com/venkatesh-spec01/diabetes-digital-twin

Research prototype trained on 100% synthetic data, checked on anonymized real data. Not medical advice.

The problem

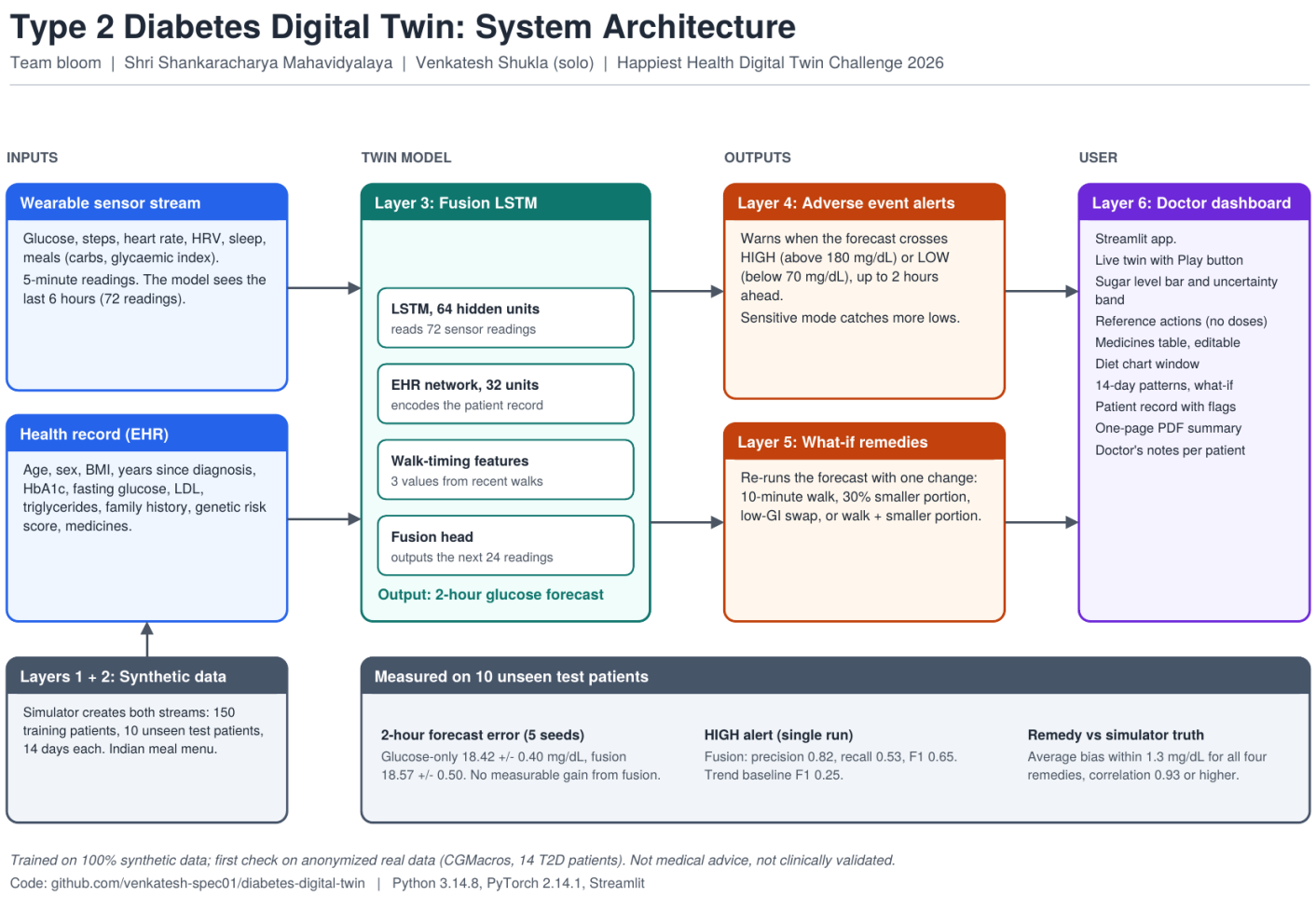
- People with Type 2 Diabetes often learn about a glucose spike only after it has happened.
- A twin that updates continuously can warn earlier, and show which small change helps most.
- Condition: Type 2 Diabetes.
- Adverse events predicted: glucose above 180 mg/dL (HIGH) and below 70 mg/dL (LOW), up to 2 hours ahead.
- Small changes tested: a 10-minute walk, a 30% smaller portion, a low-GI swap.

The solution: a six-layer twin

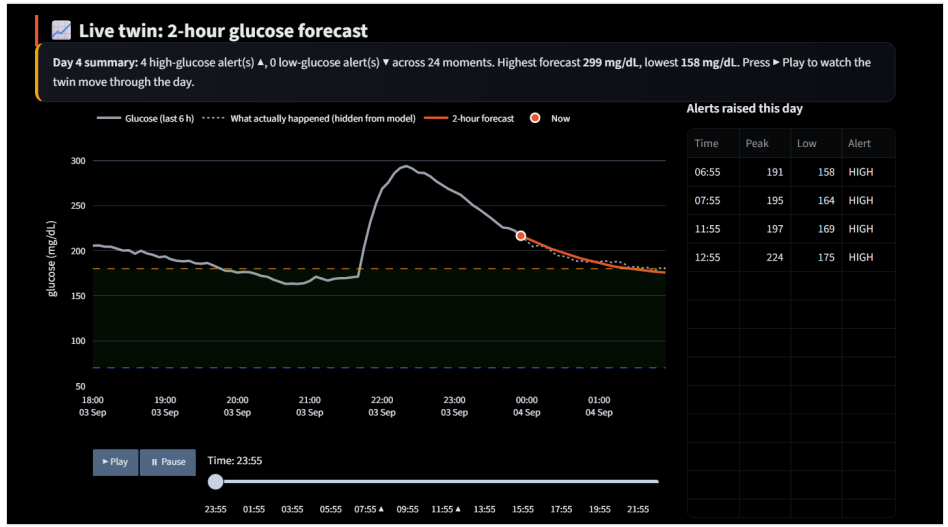
Layer	What it does
1 + 2	Synthetic EHR + wearable generator (glucose, steps, HR, HRV, sleep, meals)
3	Fusion LSTM: 6 h of sensor history + patient record, predicts next 2 h
4	Alerts for HIGH (>180) and LOW (<70) with precision / recall / F1
5	What-if remedies, checked against the simulator's true answers
6	Streamlit doctor dashboard
Check	5-seed reproducibility test

Stack: Python 3.14.8, PyTorch 2.14.1, Streamlit, Plotly. Settings: 72 readings history, 24 readings horizon, hidden 64, 12 epochs.

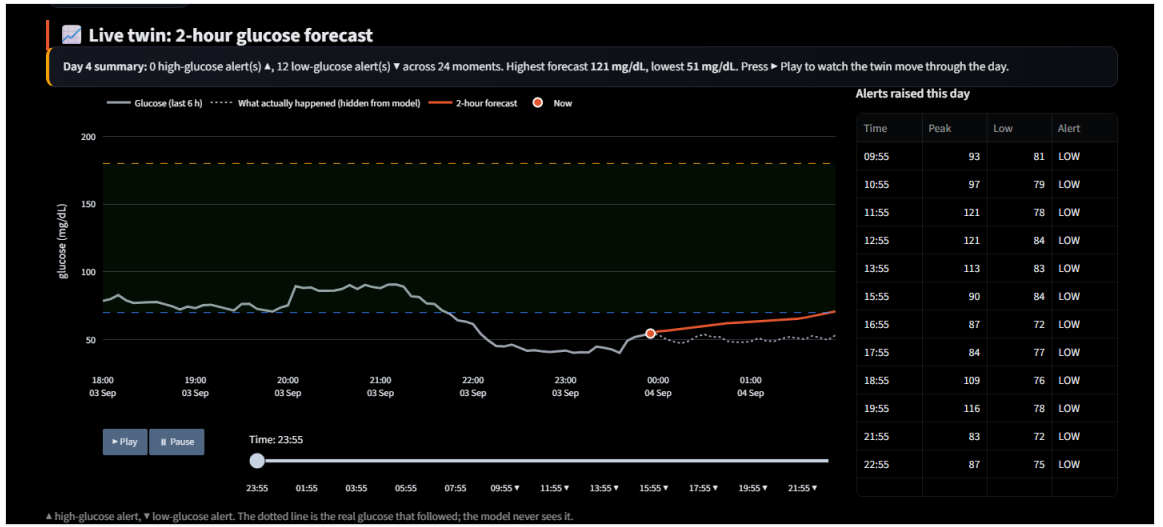
Architecture



Doctor dashboard: live twin and alerts



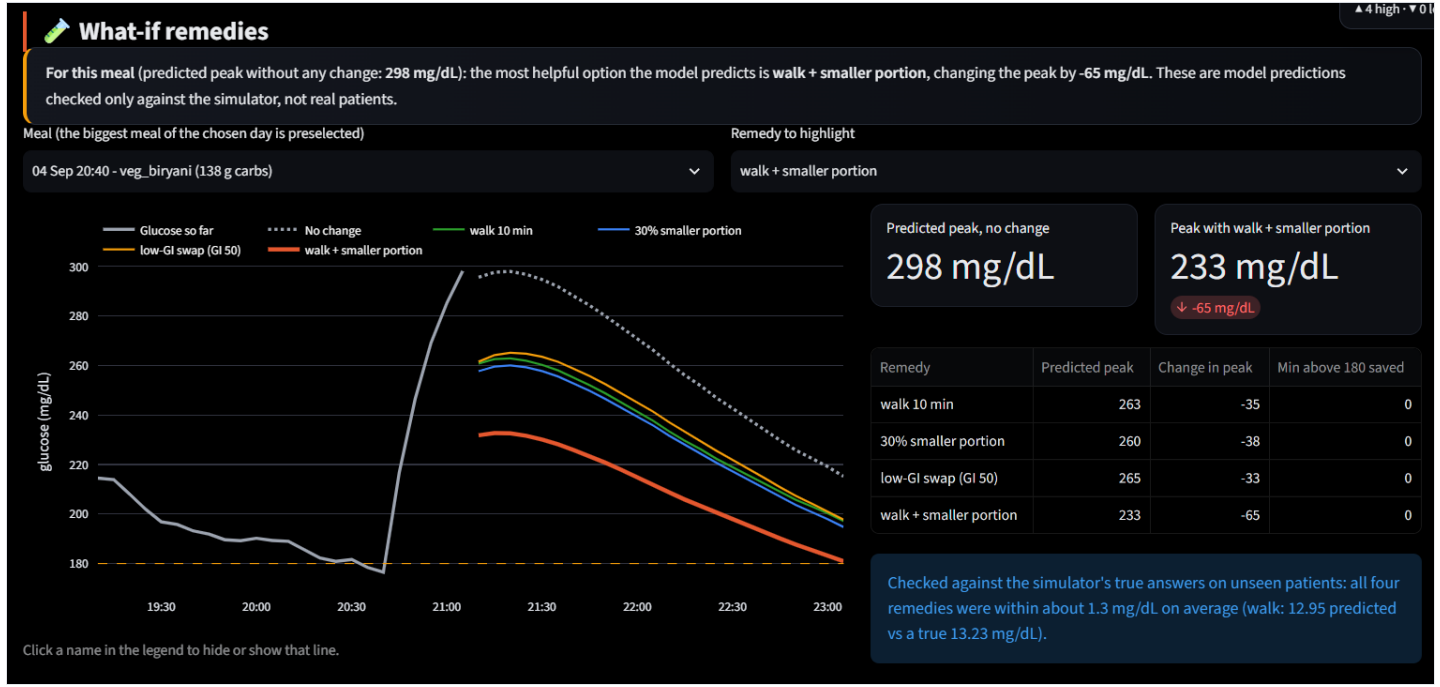
HIGH alert (T03, day 4: 4 alerts)



LOW alert (T06, day 4: 12 alerts)

- Play button replays the day; forecast appears 2 hours ahead of the real curve.
- Sensitive mode is the default because a missed low is the more dangerous error.

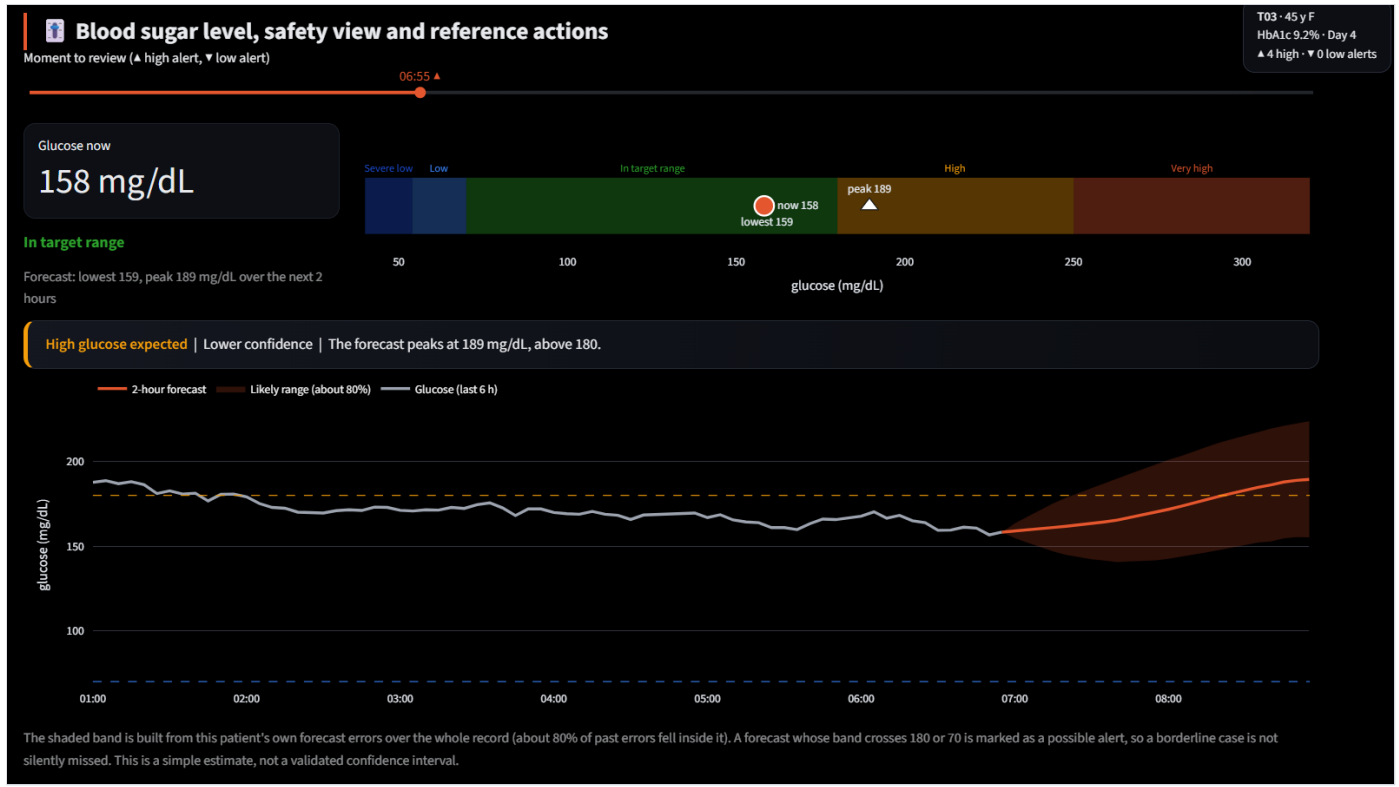
What-if remedies and 14-day patterns

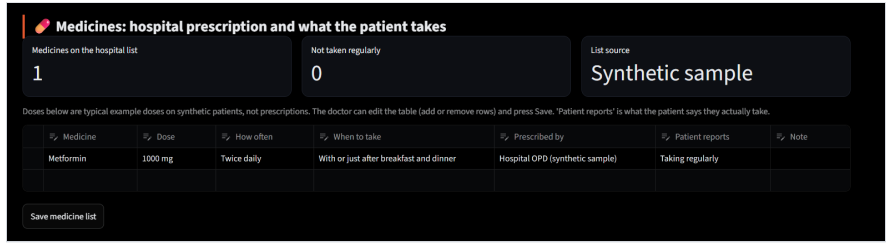




- Pick a meal, compare remedies side by side.
- T03 shown with a 138 g-carb meal, an extreme case. Average walk + smaller portion effect is about 20 mg/dL.
- T03 over 14 days: 56% in range, 44% above 180.

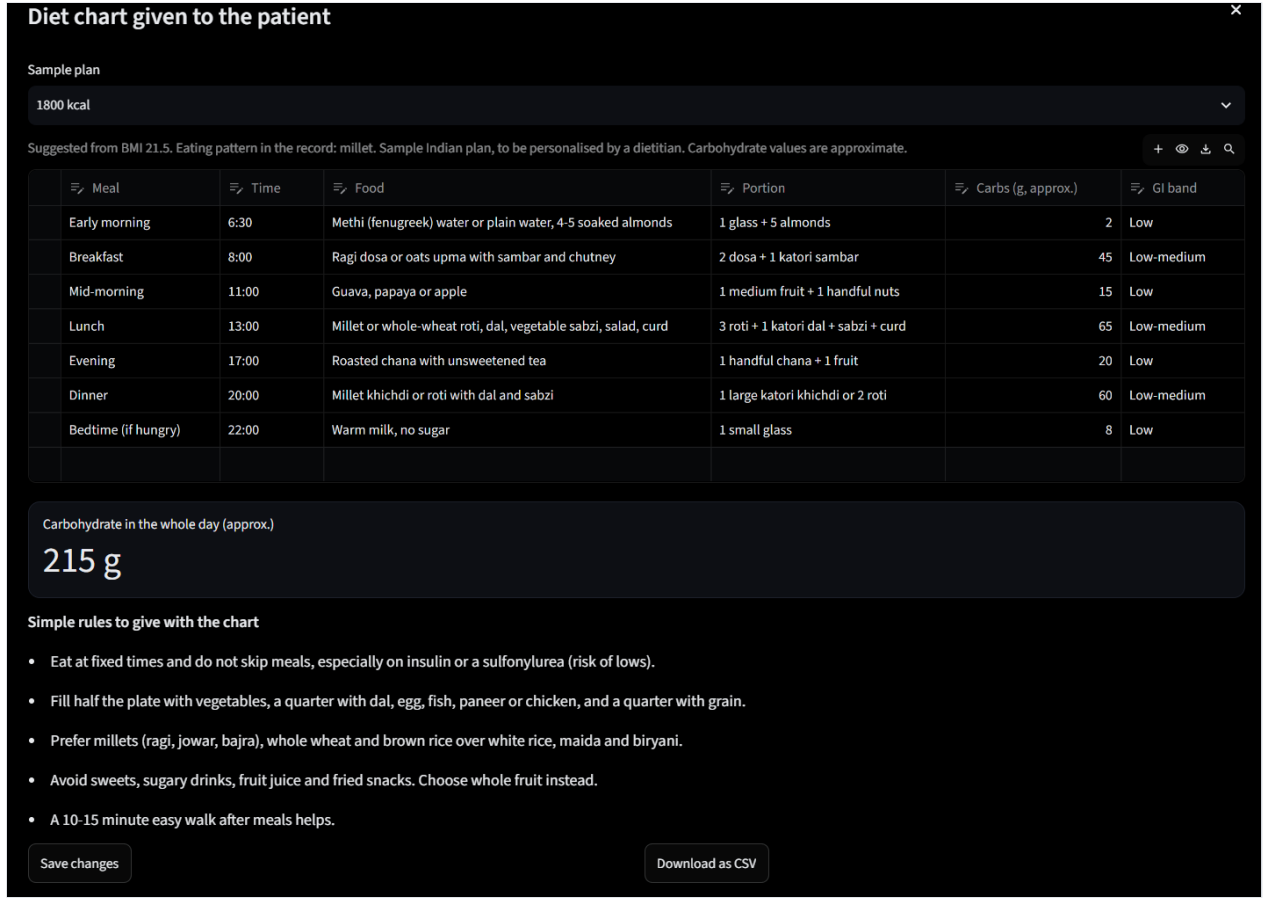
Safety view and medicines





- Coloured bar shows where glucose is now and the forecast low and peak.
- The shaded band is the likely range (about 80%). Wide band = lower confidence, flagged.
- Doctor edits the medicine list; the patient's own report sits next to it.
- Reference actions only. No doses are calculated.

Diet chart and patient summary



Diet chart given to the patient

Sample plan

1800 kcal

Suggested from BMI 21.5. Eating pattern in the record: millet. Sample Indian plan, to be personalised by a dietitian. Carbohydrate values are approximate.

Meal	Time	Food	Portion	Carbs (g, approx.)	GI band
Early morning	6:30	Methi (fenugreek) water or plain water, 4-5 soaked almonds	1 glass + 5 almonds	2	Low
Breakfast	8:00	Ragi dosa or oats upma with sambar and chutney	2 dosa + 1 katori sambar	45	Low-medium
Mid-morning	11:00	Guava, papaya or apple	1 medium fruit + 1 handful nuts	15	Low
Lunch	13:00	Millet or whole-wheat roti, dal, vegetable sabzi, salad, curd	3 roti + 1 katori dal + sabzi + curd	65	Low-medium
Evening	17:00	Roasted chana with unsweetened tea	1 handful chana + 1 fruit	20	Low
Dinner	20:00	Millet khichdi or roti with dal and sabzi	1 large katori khichdi or 2 roti	60	Low-medium
Bedtime (if hungry)	22:00	Warm milk, no sugar	1 small glass	8	Low

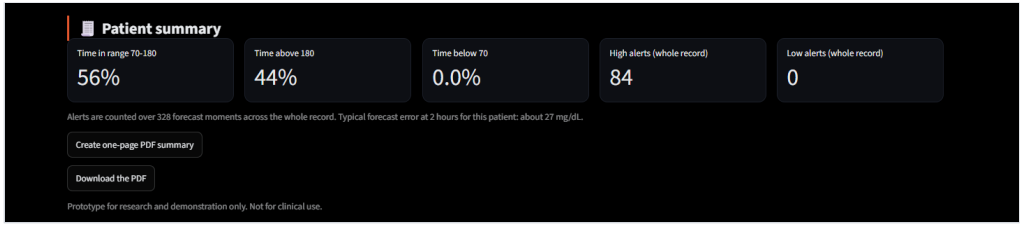
Carbohydrate in the whole day (approx.)

215 g

Simple rules to give with the chart

- Eat at fixed times and do not skip meals, especially on insulin or a sulfonylurea (risk of lows).
- Fill half the plate with vegetables, a quarter with dal, egg, fish, paneer or chicken, and a quarter with grain.
- Prefer millets (ragi, jowar, bajra), whole wheat and brown rice over white rice, maida and biryani.
- Avoid sweets, sugary drinks, fruit juice and fried snacks. Choose whole fruit instead.
- A 10-15 minute easy walk after meals helps.

Save changes | Download as CSV



Patient summary

Time in range 70-180	Time above 180	Time below 70	High alerts (whole record)	Low alerts (whole record)
56%	44%	0.0%	84	0

Alerts are counted over 328 forecast moments across the whole record. Typical forecast error at 2 hours for this patient: about 27 mg/dL.

Create one-page PDF summary

Download the PDF

Prototype for research and demonstration only. Not for clinical use.

- Diet chart opens in its own window. Sample Indian plan; the doctor edits it and saves or downloads it.
- Summary: time in range, time above 180 and below 70, alert counts.
- One click makes a one-page PDF to hand over.
- Marked as a prototype, not for clinical use.

Results on 10 unseen patients

2-hour forecast error (RMSE, 5 seeds)

Model	RMSE mg/dL
Glucose-only LSTM	18.42 +/- 0.40
Fusion LSTM	18.57 +/- 0.50

HIGH alerts: 1,661 events, 10,695 windows

Method	Precision	Recall	F1
Trend baseline	0.37	0.19	0.25
Glucose-only LSTM	0.76	0.49	0.59
Fusion LSTM	0.82	0.53	0.65
Fusion, sensitive	0.54	0.85	0.66

LOW alerts: 239 events

Method	Prec.	Recall	F1
Fusion, default	1.00	0.14	0.24
Fusion, sensitive	0.40	0.82	0.53

- Fusion is not a measurable RMSE gain (better in 2 of 5 seeds). We do not claim one.
- Alerts: fusion beats the trend baseline 0.65 vs 0.25 F1.
- Default setting: 0 false low alarms but misses most lows. Sensitive catches 82%.
- Alert numbers are a single run.

Remedies vs the simulator's true answers

Remedy	True drop	Predicted	Bias	MAE	Corr.
Walk 10 min	13.23	12.95	-0.28	1.79	0.97
30% smaller portion	11.29	11.97	+0.67	2.52	0.94
Low-GI swap (GI 50)	11.97	12.49	+0.51	2.44	0.93
Walk + smaller portion	20.42	21.69	+1.28	3.26	0.96

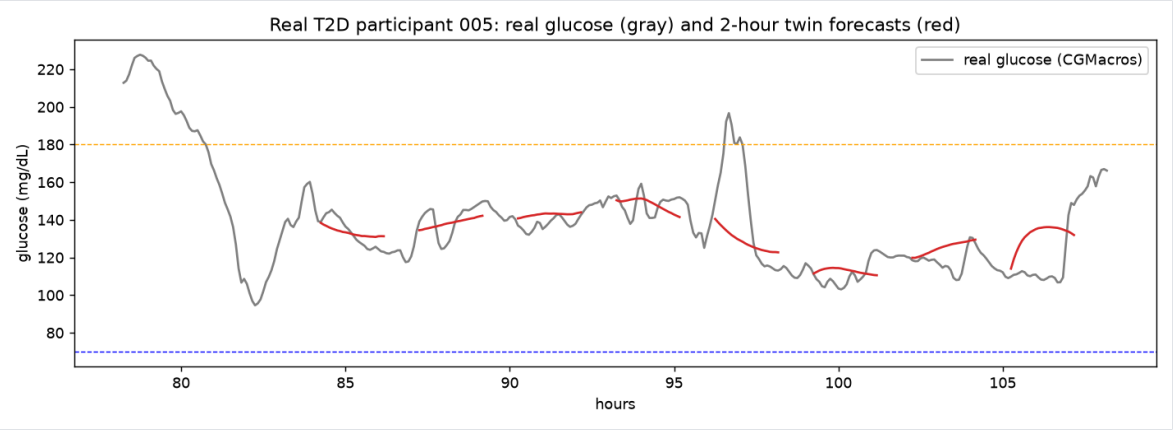
- Average bias within 1.3 mg/dL for all four remedies; correlation 0.93 or higher.
- Walk fix: the old model over-promised the walk by +1.7 mg/dL. A 75% blend of base and fine-tuned weights, chosen on validation days only, brought it to -0.3.
- Checked against the same simulator that made the data: this tests internal consistency, not real physiology.

Real-data check: CGMacros (no retraining)

45 real adults (anonymized, PhysioNet). Results for the 14 with Type 2 Diabetes, 12,884 windows.

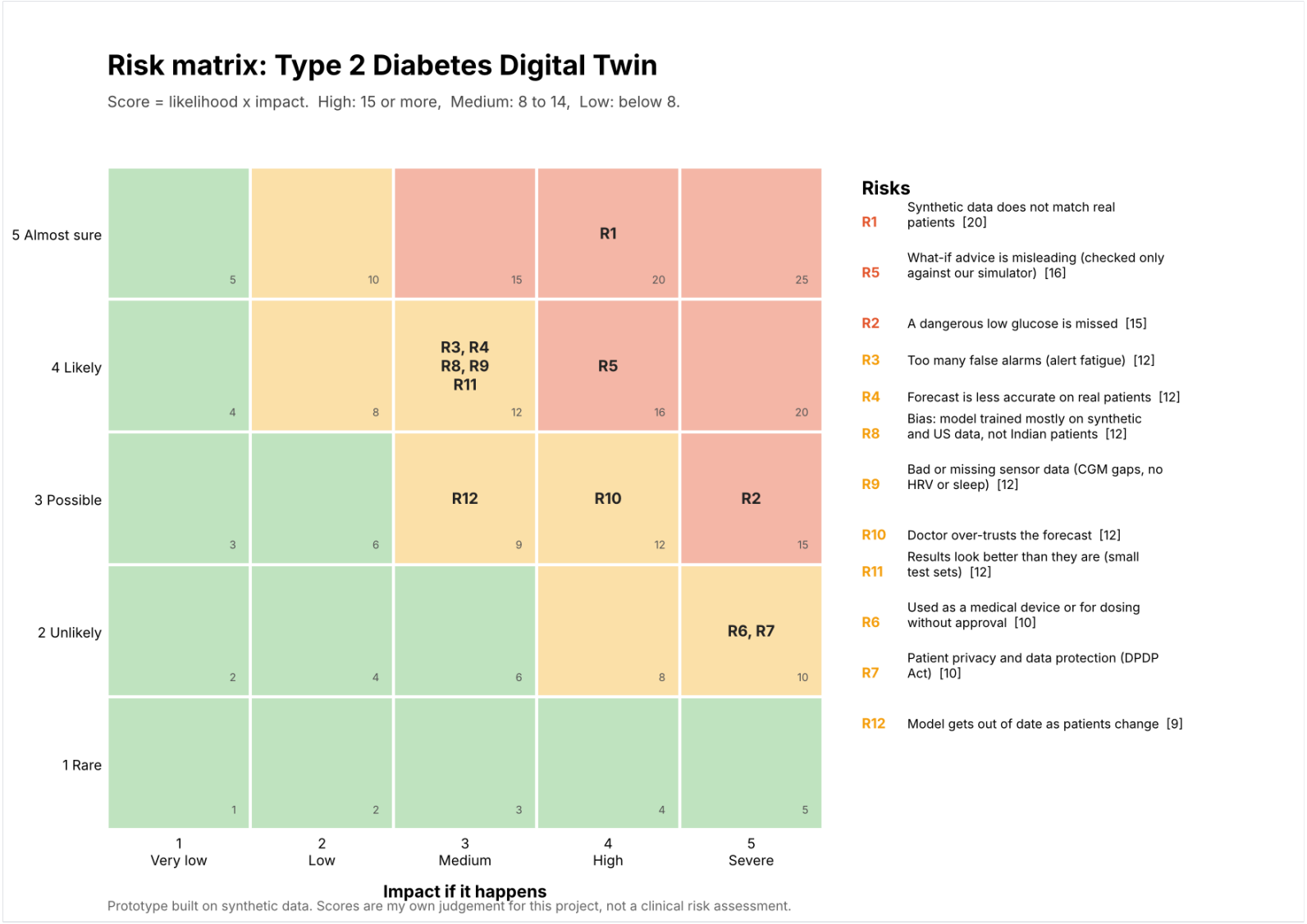
Model	30 min	60 min	120 min
No change (flat)	23.1	36.6	52.2
Trend	28.1	55.1	107.0
Sensor-only LSTM	22.4	34.3	46.1
Fusion twin	24.0	35.7	46.0

2-hour forecast error, RMSE in mg/dL (lower is better)



- Beats 'no change' at 2 h (46 vs 52), so it learned real glucose behaviour.
- Error is about 2.5x the synthetic result (18), and slightly worse than 'no change' at 30 min.
- HIGH alerts on real T2D: sensitive mode catches 37% (precision 0.57); default catches only 8%.
- Next: fine-tune on CGMacros with whole patients held out.

Risk matrix



- 12 risks scored as likelihood x impact (1 to 5 each).
- Highest: synthetic data is not real patients (20), what-if advice only checked against our simulator (16), a missed dangerous low (15).
- What we did: tested on real CGMacros data, sensitive low-alert mode, uncertainty band, wording that says the doctor decides.
- Scores are our own judgement, not a clinical risk assessment.

Regulation we researched: Indian medical device rules

- Software that monitors or forecasts glucose for a doctor counts as Software as a Medical Device under the Medical Devices Rules, 2017 (MDR 2017).
- CDSCO published its medical device software guidance in 2026. The class (A to D) depends on how serious the condition is and how much the output influences the doctor.
- Class A and B: apply on Form MD-3, licence is MD-5, from the State licensing authority.
- Class C and D: apply on Form MD-7, licence is MD-9, from the Central authority (CDSCO).
- Our read: this tool only informs a doctor, so Class A or B. If CDSCO sees it as driving decisions, B or C. CDSCO makes the final call.

How the class is worked out

Condition	Treats	Drives	Informs
Critical	D	C	B
Serious	C	B	A
Non-serious	B	A	A

Rows: how serious the condition is. Columns: what the software does with the result. Highlighted row: diabetes, which we treat as serious. Simplified from the guidance.

Where we are today

Research prototype on synthetic data. No licence applied for. Every screen says not for clinical use.

Standards and data rules we designed around

Rule	What it asks for	What we did
ISO 13485	A documented quality system in the company	Build order table in the README, pinned versions, full code history. The company-level system comes later.
IEC 62304	A controlled software lifecycle: requirements, tests, releases	Each script does one job, fixed seeds, 5-seed repeat test, models and data committed so results can be reproduced.
ISO 14971	A risk management file	12-risk matrix with likelihood, impact and what we did. A first draft of the file.
DPDP Act	Consent, only the data needed, security	Synthetic patients and anonymised open data (CGMacros). Nothing that identifies a person.
CDSCO MDR 2017	A licence based on class	Prototype wording on every screen, reference actions only, no doses calculated.
ABDM / ABHA	Optional link to hospital health records	Not built. A later step after the licence and a hospital pilot.

Our own code is MIT licensed. The standards are listed in CDSCO guidance as applicable; we followed their ideas but have not been audited against them.

Limitations and next steps

Limitations

- Trained on synthetic data only; real-data error is 2.5x larger.
- Remedy checks use the same simulator.
- Fusion gives no 2-hour RMSE gain.
- Walk + portion still slightly optimistic (+1.3).
- Alert numbers from a single run.
- Not a medical device.

Validation plan

- Fine-tune on CGMacros with held-out patients.
- Add prediction intervals and Clarke Error Grid.
- Tune thresholds on a separate validation set.
- Report alerts per episode and warning time.
- Clinician pilot under ethics approval.

Outcome

- Working end-to-end twin: data, forecasting, alerts, remedies, dashboard.
- Fusion LSTM raises HIGH-alert F1 from 0.25 (trend) to 0.65.
- Remedy predictions track the simulator's truth within 1.3 mg/dL on average.
- On 14 real T2D patients, with no retraining, it beats 'no change' at 2 h (46 vs 52 mg/dL).
- Honest reporting: where fusion does not help, we say so.
- Open source (MIT). Reproducible: pinned requirements, seeds, committed data and models.
- Code: github.com/venkatesh-spec01/diabetes-digital-twin